



Manufacturing change management – an AI- and data-enhanced Delphi study and algorithm to support change process tailoring and the identification of suitable methods and digital tools

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Abstract

Today's manufacturing industry is exposed to increasing external and internal disturbing issues, including supply chain disruptions, geopolitical uncertainties, and workforce shortages. These dynamics require companies to frequently implement Manufacturing Changes (MCs), coordinated through structured Manufacturing Change Management (MCM) processes. However, existing MCM approaches often fall short in adapting process steps and selecting suitable support tools to the specific characteristics of a given change. This results in inefficiencies and limited responsiveness. To address this gap, this paper develops the core logic for a change-specific and company-individual MCM methodology. The focus lies on establishing and operationalizing correlations between change characteristics, process logic, and supporting methods and digital tools (M&DTs). A three-phase approach was applied: a hybrid Delphi study combined expert input and artificial-intelligence-supported assessments to derive initial correlation matrices, then selected dependencies were verified with real-world change data, and a configurable algorithm was developed to transform structured input into tailored change processes and M&DTs recommendations. The resulting framework and application methodology enables manufacturing companies to assess MCs based on structured attributes and to derive adapted process models, including suitable M&DTs. This contributes to a more efficient and context-specific handling of changes in dynamic industrial environments.

Keywords Change management · Process · Flexibility · Methods · Digital tools · Delphi study · Algorithm

1 Introduction

Manufacturing companies today face an increasingly volatile environment to which they must continuously adapt [1]. In particular, external factors like geopolitical tensions, trade conflicts [2], and recurring supply chain disruptions [3] have significantly impacted their operational business in recent years. In addition, internal factors such as a high staff turnover [4] or shortages of skilled workers [5] challenge companies to adapt to these conditions and maintain a certain level of operational flexibility. Beyond these factors,

modern manufacturing is increasingly characterized by high product and process variability driven by mass customization and dynamic market requirements. This increases the frequency and complexity of manufacturing changes and calls for flexible and adaptive coordination mechanisms, particularly in human–robot collaborative and small-batch production environments [6, 7]. At the same time, the growing use of collaborative and reconfigurable robotic systems enhances flexibility but introduces additional interdependencies between processes, technologies, and organizational structures, reinforcing the need for context-aware change management approaches [6, 7].

For manufacturing companies, this means that they must regularly implement changes to their production system, known as Manufacturing Changes (MCs). In the literature, MCs are defined as “any alteration made to the factory or its elements” [8]. Accordingly, both minor changes, such as adjustments of machine parameters, and major changes, such as redesigning a

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factory layout, fall under this definition. The handling of these changes is referred to as Manufacturing Change Management (MCM) [8] and constitutes the scientific frame of reference for this contribution. Studies, such as the one by Rammo and Graf [9], show that MCs are increasing in frequency and complexity. Likewise, McKinsey and Company (2018) [10] reports that unmanaged change processes can cause significant inefficiencies and cost overruns in manufacturing.

The efficient management of MCs is, therefore, essential for the competitiveness of manufacturing companies. However, the diversity of MCs introduces a major challenge. Each change type entails different parameters and requirements, such as the number of necessary process steps [11] or the need for supporting methods and digital tools [12]. For instance, a recurring minor adjustment, such as changing a machine parameter, differs substantially from a structural change like reconfiguring a production line. The latter may benefit from additional planning stages and creativity methods, while the former might require only minimal intervention.

In industrial practice, MCM processes are often too rigid [11] and lack sufficient methodological and digital support to cope with this wide range of changes [13]. This limited adaptability reduces efficiency and effectiveness, leading to higher efforts and ultimately financial losses [14]. Current approaches in MCM research only marginally address the need for flexibility and for an appropriate methodological and digital support of change processes. The few approaches that do consider these aspects focus exclusively on generic MCM processes. This neglects the fact that MCM processes can differ significantly between companies, for example, in terms of the number of process steps, the involved stakeholders, or the required documents.

To address this challenge, Rammo and Zaeh [15] propose an approach for a change-specific and company-individual MCM. The approach comprises four main steps (Fig. 1): the process description (Step 1), the change description (Step 2), the identification of methods and digital tools (M&DTs) (Step 3), and finally, the correlation (Step 4). A comprehensive application methodology supports potential users throughout all these steps. It enables manufacturing companies to flexibly define the necessary process steps as well as the suitable M&DTs for particular changes based on their individual MCM processes.

This paper focuses on Step 4, the correlation, which represents the core element of the methodology. In this step, the dimensions developed in previous publications, i.e., the change process description [16], the change description [17], and the M&DTs catalog and description [18], are correlated with each other based on their respective attributes. For example, the attribute *complexity* of a change description can determine that more analysis steps are required in the change process and that more elaborate M&DTs become relevant. Consequently, the aim of this contribution is to develop a correlation matrix as well as an algorithm for the overall methodology, which derives the required process steps and suitable M&DTs from the assessment of a given change. As a theoretical contribution, this work expands the knowledge about the relationships between changes, process steps, and the use of M&DTs. Furthermore, by introducing the algorithm, it establishes a systematic way to connect these elements in a consistent and reproducible manner. As a practical contribution, this paper provides a user-oriented application methodology that enables practitioners to assess each upcoming change and automatically derive a tailored

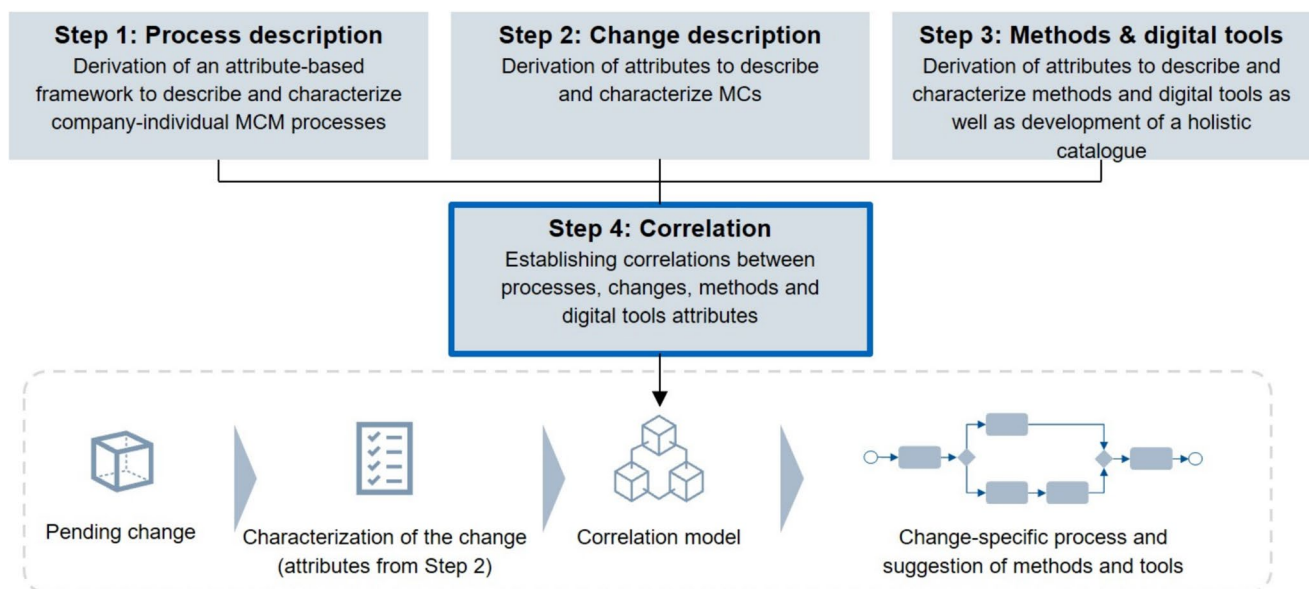


Fig. 1 Integration of this contribution into the overall methodology

process, comprising the relevant process steps and appropriate supporting M&DTs. This increases the efficiency and effectiveness of implementing changes while also providing a structured basis for the consistent collection of process data and documentation.

The remainder of this contribution is structured as follows. Section 2 reviews the current state of research in MCM and related fields, identifying key limitations and the resulting research gap. Section 3 outlines the overall methodology and its three-phase structure. Section 4 presents the detailed procedure and results of each phase, including the hybrid Delphi study, the data-based verification, and the algorithm development. Section 5 discusses the findings in light of the practical application, the implementation challenges, and the conceptual limitations. Finally, Sect. 6 concludes the contribution and gives an outlook on future research.

2 State of the art

The literature review conducted for this section followed the methodology proposed by Vernim et al. [19], which involves developing a concept map with initial keyword clusters that are iteratively refined to improve the search results (Fig. 2). For this study, the keyword development and the search strategy were conceptually divided into two thematic areas based on the following questions:

1. **Application domain:** Has prior research in MCM or Engineering Change Management (ECM) already examined the interdependencies between changes, change processes, and M&DTs? How have they approached this research?
2. **Methodology/approach:** How have other fields approached this topic? Which methods or frameworks have they used to elaborate such interdependencies?

The respective keywords were applied in both English and German to account for relevant literature in both languages and were combined using Boolean operators to refine the search strings. The search covered the period from 2014 to 2025 and was conducted across the scientific databases Scopus, ScienceDirect, and Web of Science. The relevance was assessed through a multi-step screening process, beginning with a review of the titles and the abstracts, followed by the full texts to confirm the thematic alignment with the research objectives. From the originally identified 221 publications, 21 were ultimately deemed relevant for this contribution.

Eight of the 21 identified publications address interdependencies between changes, change processes, and M&DTs. These works can be grouped into two clusters: approaches focusing on flexible adaptation of change processes and approaches emphasizing the integration of M&DTs into MCM.

In the first cluster, Haghi et al. [11] propose a methodology that derives a reference process and adapts it through flexibility factors linked to change attributes identified in expert

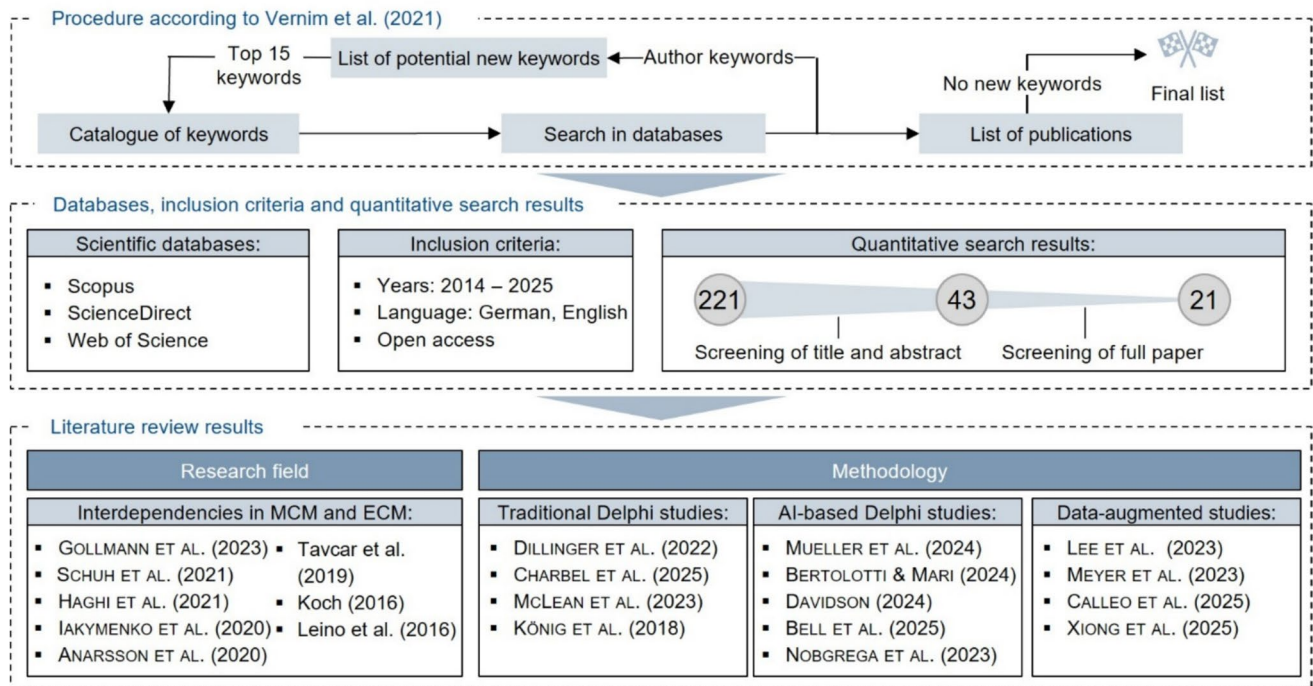


Fig. 2 Overview of the literature review procedure and the results

workshops and structured process modeling. Koch [8] similarly presents a process-based MCM approach with MC-specific adaptations derived from literature, case studies, surveys, and workshops. Changes are categorized into a small number of classes, each mapped to a corresponding process path.

The second cluster emphasizes the integration of M&DTs into MCM. Leino et al. [20] apply action research to introduce a case study focusing on digital manufacturing and Product Lifecycle Management (PLM) configurations and co-develop process models and PLM impact analyses with company stakeholders. Gollmann et al. [21] examine interdependencies between organizational structures, IT systems, and processes through expert interviews in a medical devices company. Iakymenko et al. [22] conduct multiple case studies showing how company-specific constraints, such as low production volumes or high customization, restrict ECM tool use and consequently shape process execution and change propagation.

Data-driven approaches also play a role. Schuh et al. [23] link engineering change descriptors to manufacturing operations using historical change data and simulation to quantify process impacts. Arnarsson et al. [22, 24] analyze change request logs using Design Structure Matrices (DSM) and Markov chains to identify recurring process patterns and bottlenecks and use these insights to guide process improvements. Zavbi et al. [25] embed a five-step knowledge management model into the ECM process and map activities such as change impact analysis and tacit knowledge capture to specific process stages, validated through expert assessment.

Overall, existing contributions establish interdependencies mainly through expert assessments, predefined change categories, and case-specific modeling. Most approaches focus on individual dimensions, either process flexibility or M&DTs integration, rather than capturing their combined interaction. Furthermore, many rely on static process variants and descriptive tool integration without formalized, logic-based decision support. As a result, no existing approach provides a configurable linkage between change attributes, company-individual process steps, and M&DTs with sufficient detail and adaptability. This gap motivates the methodology developed in this contribution.

Adjacent fields offer methodological insights into deriving and validating complex inter-dependencies. Three groups can be distinguished: classical Delphi studies, Generative-AI-supported Delphi formats, and data-augmented Delphi approaches.

Classical Delphi studies remain widely used for structured expert elicitation and consensus-building. Examples include Dillinger et al. [26], Charbel et al. [27], McLean et al. [28], and Koenig et al. [29], who apply Delphi designs to risk modeling, stakeholder prioritization, continuous improvement, and process deviation analysis. These works

demonstrate that iterative feedback loops can yield stable consensus on complex system relationships.

More recent studies integrate Generative AI into Delphi processes. Mueller et al. [30], Davidson [31], Bertolotti & Mari [32], Bell et al. [33] and Nóbrega et al. [34] show that AI agents can function as synthetic experts or moderators, improving consistency, broadening perspectives, and reducing iteration cycles while preserving methodological rigor.

Data-augmented Delphi designs further combine expert judgment with empirical data. Lee et al. [35], Meyer et al. [36], Calleo et al. [37], and Xiong et al. [38] demonstrate how operational and simulation data can validate expert input, mitigate bias, and identify non-obvious correlations. These hybrid formats enhance robustness and adaptability, particularly in complex decision environments.

Overall, hybrid approaches that blend expert input, AI support, and empirical data have proven effective in deriving reliable interdependencies. Building on these insights, this contribution adopts a hybrid Delphi design with complementary data-based verification to develop a flexible, logic-based methodology for a change-specific and company-individual MCM.

3 Methodical approach

The methodical approach applied in this contribution consists of three sequential phases (Fig. 3), each building on the previous phase to create a structured and operationalizable model for linking change characteristics with process steps and supporting M&DTs.

In Phase 1, a hybrid Delphi study was conducted to establish initial correlation matrices between change attributes, process attributes, and M&DTs attributes. This phase included three parallel streams: a conventional expert-based Delphi study, a GenAI-based Delphi study using fine-tuned agent personas, and a hybrid version that integrated both perspectives. In this context, personas refer to systematically designed, role-specific agent profiles that emulate the perspectives, expertise, and decision-making patterns of different stakeholder types involved in change processes. The resulting matrices from the hybrid study served as the basis for subsequent validation.

In Phase 2, selected correlations from the initial matrix were empirically verified using real-world industrial data. This data-driven validation focused on observable links between types of changes and executed process steps, refining the correlation matrix based on statistically derived patterns.

In Phase 3, the final matrices and input structures were integrated into a computational algorithm. This algorithm transforms user-specific input, such as change descriptions and company-individual process models, into tailored change process workflows.

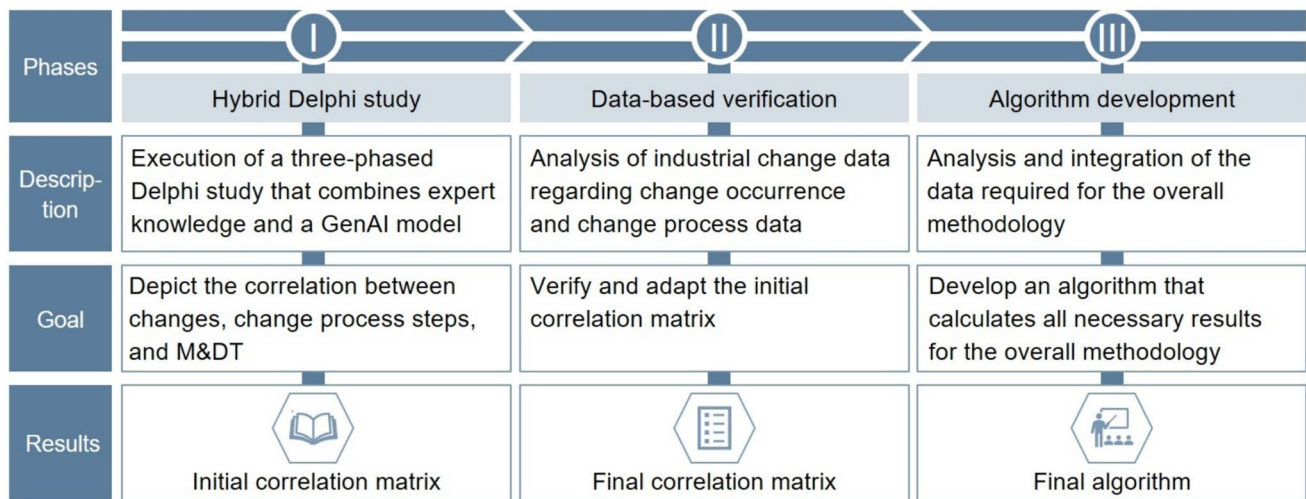
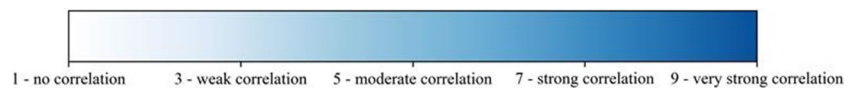


Fig. 3 Methodical approach of this paper

Fig. 4 Scale for the correlation matrices



4 Procedure and results

This chapter outlines the execution of the methodical approach in three phases: the hybrid Delphi study (4.1), the data-based verification (4.2), and the algorithm development (4.3). Each step builds on the previous to derive, validate, and operationalize the interdependencies between changes, processes, and M&DTs (Fig. 4).

4.1 Hybrid Delphi study

In the first phase, a hybrid Delphi study was conducted to identify and quantify inter-dependencies between process attributes (PAs), change attributes (CAs), and M&DTs attributes (MAs/DTAs). Based on insights from the literature (cf. Chapter 2), the study was designed to combine expert knowledge with GenAI support in a structured, multi-stage process. Figure 5 summarizes the overall design.

The study consisted of three parallel streams: First, a fully expert-based Delphi study, second, a GenAI-based Delphi study using AI agents fine-tuned with industrial change data, and third, a hybrid version that integrated results from both sources. This setup enabled a comparative evaluation of the approaches and served as the basis for selecting the final correlation matrix used in the algorithm. The different streams are detailed below.

4.1.1 Expert-based Delphi study

The expert-based Delphi study followed a structured design by Haeder [39]. First, the study objectives were operationalized through a set of correlation matrices linking CAs, PAs, and MAs/DTAs. These were embedded in a digital, Excel-based questionnaire, supported by definitions, examples, and explanations for each attribute. The experts were asked to assess the strength of the correlation between attribute pairs using a 5-point Likert scale, where 1 indicates no correlation and 9 a very strong correlation. The scale consisted only of odd values (1, 3, 5, 7, 9), to ensure a clearer differentiation between response options through wider intervals. The scale for all three figures is shown in Fig. 4. To illustrate the rating logic, selected attribute relationships are briefly exemplified. For instance, the change attribute *complexity* was assessed in relation to the process attribute *analysis effort*. A high rating (e.g., 7) reflects the assumption that more complex changes typically require more extensive analytical activities. Similarly, the process attribute *analysis effort* was strongly correlated with the method attribute *analysis support* (e.g., rating of 9), as analytical methods are directly intended to support such process steps. A consensus threshold was defined based on the percentage (more than 75%) [40] of experts whose ratings deviated no more than ± 2 from the median.

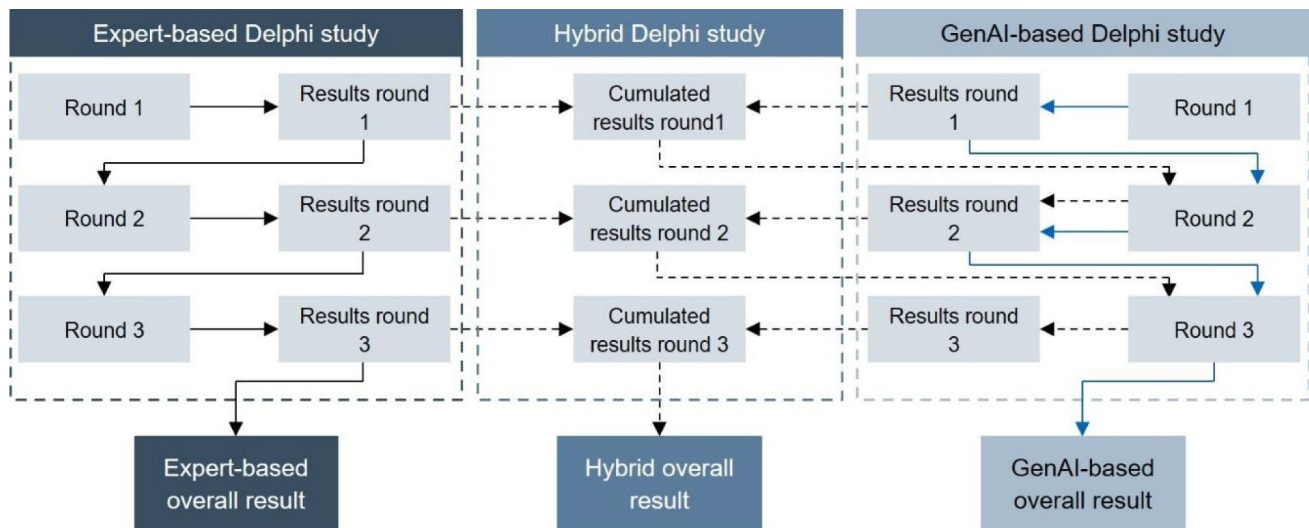


Fig. 5 Procedure of the hybrid Delphi study

The panel consisted of 18 experts with at least five years of experience in manufacturing or change-related roles. A pretest ensured the usability of the questionnaire. The Delphi process was conducted in three rounds. After each round, the experts received anonymized feedback on the group median, interquartile range, the current consensus, and their individual ratings. In round 3, consensus was reached across all matrices, and the results served as the output of the expert-based stream.

4.1.2 GenAI-based Delphi study

In parallel, a GenAI-based Delphi study was conducted using 18 virtual expert agents based on OpenAI's GPT-4o model. Each agent was assigned a persona reflecting a realistic expert profile, including background, role, personality traits, and technical knowledge. The agents were fine-tuned using real-world industrial change data, encompassing historical change cases, corresponding process executions, and contextual metadata. The GenAI Delphi study followed the same structure as the expert study, with the AI agents independently completing all three rounds. Prompts were structured according to the best practices from Sahoo et al. [41] to maximize consistency and realism of the agent responses. To ensure consistent behavior of the AI-generated personas across Delphi rounds, persona definitions, role descriptions, and prompt structures were kept constant throughout the study. Consistency was monitored by comparing rating distributions and consensus development across rounds, which showed stable and coherent progression patterns comparable to those observed in the expert-based Delphi stream. In addition, internal consistency was assessed by checking for coherent rating patterns across related attributes within each

persona and the absence of contradictory assessments. The consensus threshold was identical to the expert-based study.

4.1.3 Hybrid Delphi study

The hybrid study integrated the experts' and the GenAI results in a combined panel of 36 participants. Both groups' responses were weighted equally. In round 2, only the AI agents received the aggregated feedback from all 36 participants. The real-world experts remained uninfluenced by the AI results to preserve independence and avoid bias, aligning with recommended practices in hybrid Delphi research. Round 3 again aggregated responses from both sources, producing the final hybrid correlation matrices.

Comparative analysis showed that the GenAI agents reached higher consensus levels more quickly than the human experts, particularly in early rounds. However, the experts added valuable contextual interpretations. The hybrid study effectively combined the consistency and scalability of AI with the domain-specific judgment of the experts.

The final results of this phase are the correlation matrices derived from the hybrid Delphi study, which capture the perceived interdependencies across all three domains (CA, PA, MA/DTA). These matrices form the foundation for the next phase, in which selected relationships were empirically verified using real-world change data.

4.2 Data-based verification

In the second phase, the correlation matrices resulting from the hybrid Delphi study served as the foundation for a data-based verification. Since real-world data was only available for changes and process steps, but not for associated

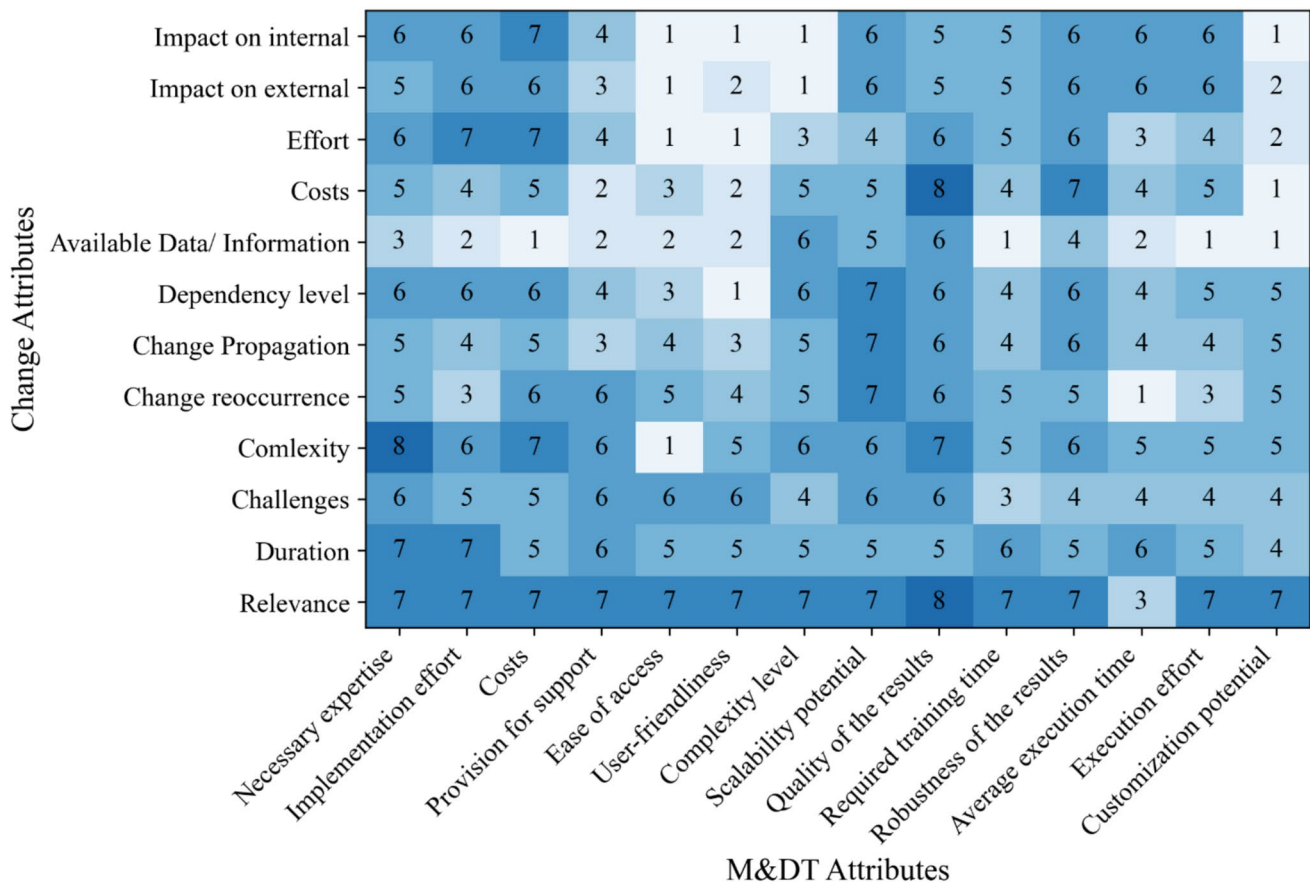


Fig. 6 Matrix 1 of the Delphi study, showing the correlation of CAs and MAs/DTAs

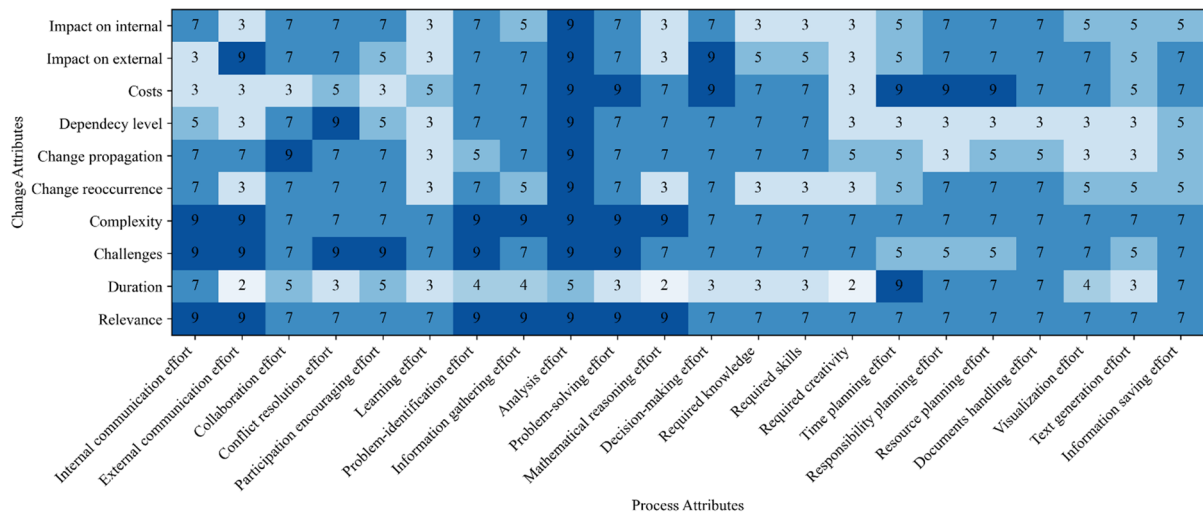


Fig. 7 Matrix 2 of the Delphi study, showing the correlation of CAs and PAs

M&DTs, this verification focused on confirming a subset of the interdependencies identified in the initial matrices. The objective was to examine which types of changes, described by defined change attributes, lead to which process steps in actual practice. To achieve this, an industrial dataset

with more than 3000 changes was used that contained both records of individual MCs and the corresponding process steps executed in response. The process steps were manually characterized using the process attributes developed by Rammo et al. [16]. Similarly, the changes were categorized

Process Attributes	Internal communication effort	1	1	9	3	2	3	2	1	3	5
	External communication effort	1	1	8	3	2	2	2	1	3	5
	Collaboration effort	2	3	9	3	3	3	2	1	3	5
	Conflict resolution effort	1	1	7	2	1	1	2	3	2	3
	Participation encouraging effort	1	1	8	1	3	3	3	5	2	2
	Learning effort	1	1	3	1	3	3	4	8	1	3
	Problem-identification effort	3	7	1	2	3	2	2	1	1	4
	Information gathering effort	5	5	3	3	1	2	1	1	1	5
	Analysis effort	3	9	3	1	3	2	1	1	2	6
	Problem-solving effort	3	7	3	2	5	3	2	1	3	6
	Mathematical reasoning effort	3	7	3	1	5	1	2	2	2	4
	Decision-making effort	4	7	5	3	6	3	3	1	2	6
	Required knowledge	5	1	2	1	4	5	6	1	5	5
	Required skills	4	1	2	2	6	5	5	3	4	5
	Required creativity	2	1	4	1	3	2	9	4	1	5
	Time planning effort	6	3	2	2	3	4	2	1	8	6
	Responsibility planning effort	5	2	2	1	3	4	2	2	8	4
	Resource planning effort	6	3	2	1	3	4	2	1	8	4
	Documents handling effort	5	2	1	2	4	9	1	1	2	1
	Visualization effort	4	4	3	3	1	7	3	1	2	9
	Text generation effort	5	2	3	2	2	7	3	1	1	3
	Information saving effort	5	3	1	1	2	7	1	1	3	3
		Automation/integration	Analysis/evaluation	Collaboration/communication	Controlling/monitoring	Deployment/execution	Documentation	Ideation/development	Personal development/training	Planning	Visualization
		M&DT Attributes									

Fig. 8 Matrix 3 of the Delphi study, showing the correlation of PAs and MAs/DTAs

using the change attributes developed by Rammo et al. [17]. Instead of assigning attributes individually to every change in the dataset, twenty predefined change categories were used, each with fixed attribute values, allowing for a scalable and consistent classification for the large number of changes in the dataset.

From this structured data, statistical correlations were calculated between change attributes and process attributes. This enabled an empirical identification of interdependencies between types of changes and the steps typically triggered in the change process. In this way, parts of the Delphi-based correlation matrix could be verified using industrial data.

The resulting correlation values were then normalized to match the scale used in the Delphi study. A comparison was conducted to identify potential deviations between the expert-based and data-based results. Overall, only a few deviations were observed, and where necessary, adjustments were limited to a one-level shift on the defined correlation scale.

As a result, this phase produced refined correlation matrices that combine the Delphi insights with empirical verification. Figure 6 shows the final result for CAs and MAs/DTAs, Fig. 7 for CAs and PAs, and Fig. 8 for PAs and MAs/DTAs.

4.3 Algorithm development

In the third phase, an algorithm was developed that serves as the computational core of the overall methodology. Its purpose is to process and combine all relevant data to generate the final output: a process model adapted to the specific characteristics of a given change, based on the company-individual change process and enriched with tailored recommendations for M&DTs. The algorithm processes all relevant inputs: predefined structural data (such as correlation matrices and attribute profiles), initial user selections (such as process models and threshold settings), and operational input that describes a specific change scenario. Before introducing the mathematical formulation of the algorithm, it is helpful to first present the conceptual structure of the methodology from a user's perspective. This practical view illustrates how the algorithm is embedded in the overall process logic and how the method is applied in real-world scenarios.

The application methodology (Fig. 9) is divided into two segments: an initialization phase, which is carried out once per company, and an operational phase, which is executed for each new change case.

The initialization phase consists of three application steps. First, the company-individual change process is formalized and integrated into the method. This is achieved using a Design Structure Matrix (DSM), which captures the sequence and dependencies between individual process steps in a binary matrix. The DSM enables the algorithm

to process complex workflows and preserve the execution logic. Each process step in the DSM is then described using process attributes, based on the attribute framework proposed by Rammo et al. [16], capturing key characteristics such as *analysis effort* or *communication effort*. Second, the user defines which change attributes will be used to describe future change scenarios. Based on Rammo et al. [17], these may include, e.g., *complexity* or *implementation effort*. The selected attributes define the structure by which each change is evaluated. Third, the user specifies the M&DTs that should be considered within the configuration, selecting from a catalog in which relevant M&DTs are described, or can add company-specific M&DTs using a standardized attribute set following Rammo et al. [18].

Once initialized, the methodology enters the operational phase, in which it is applied to individual change cases. For each new change, the user provides an attribute-based evaluation. Based on this input, the algorithm calculates which process steps from the predefined DSM are relevant and determines which M&DTs best support those steps. The result is a tailored change process model, visualized as a workflow that includes only the relevant steps, each enriched with context-specific M&DTs recommendations.

With the application methodology now outlined, the following section introduces the change-specific and company-individual Manufacturing Change Management (CCMCM) algorithm, which systematically combines all structured inputs, primarily in matrix and vector form, to generate the final initialization result.

The presentation of the algorithm begins with a definition of all required input structures, which are mathematically formalized and positioned within the overall methodology. In the first application step, a DSM is used to represent the company-individual change process. The DSM ($dsm_process \in \{0,1\}^{p \times p}$) captures the interdependencies between individual process steps (PS); a value of 1 indicates a direct relationship between two steps. The arrow notation used throughout this work, such as $PS_1 \rightarrow PS_2$, denotes directional influence, in this case, describing how one process step relates to or impacts another:

$$dsm_process = \begin{bmatrix} PS_1 \rightarrow PS_1 & \cdots & PS_1 \rightarrow PS_p \\ \vdots & \ddots & \vdots \\ PS_p \rightarrow PS_1 & \cdots & PS_p \rightarrow PS_p \end{bmatrix} \quad (4.1)$$

Also, in the first step, each PS is described using a set of process attributes (PAs). These attributes are assessed using a 5-point Likert scale ranging from 1 to 9, capturing the intensity or relevance of each attribute per step. This information is formalized in the following attribute matrix ($dmm_process \in \{1,9\}^{p \times m}$):

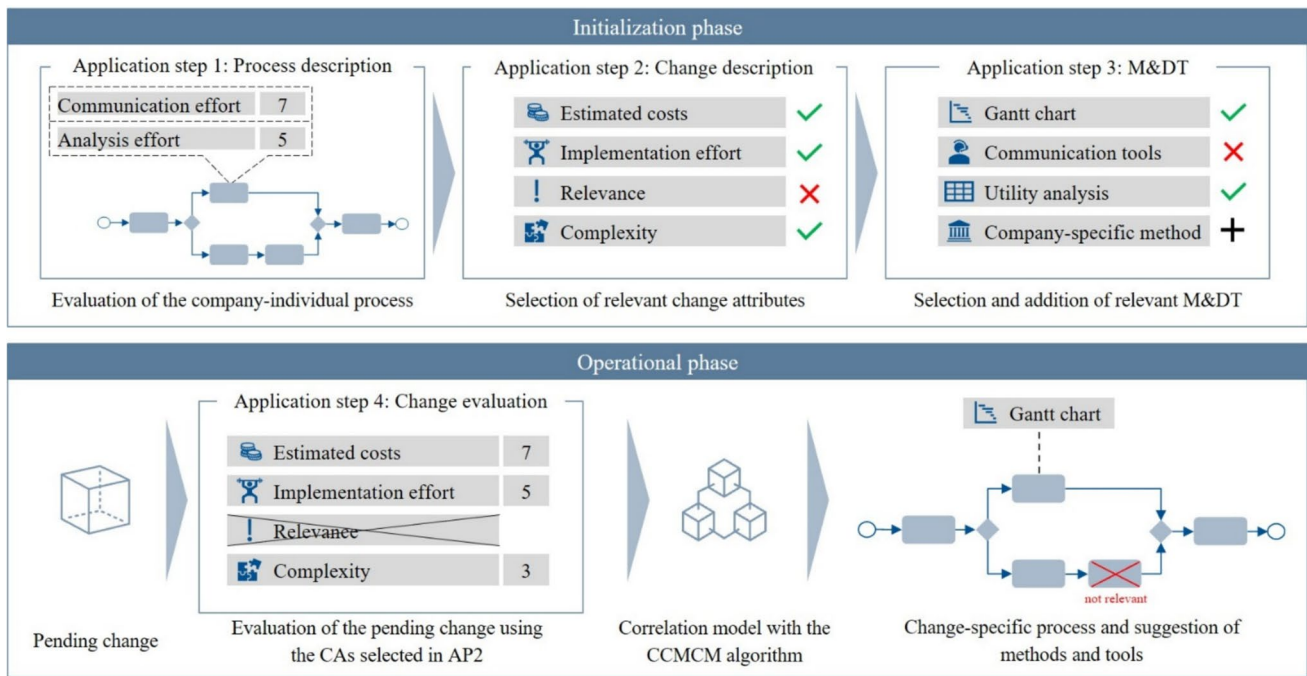


Fig. 9 Application method of the overall methodology

$$dmm_process = \begin{bmatrix} PS_1 \rightarrow PA_1 & \cdots & PS_1 \rightarrow PA_m \\ \vdots & \ddots & \vdots \\ PS_p \rightarrow PA_1 & \cdots & PS_p \rightarrow PA_m \end{bmatrix} \quad (4.2)$$

In the second application step, relevant change attributes (CAs) are selected based on the company's preferences. This selection is represented as a binary change vector ($change_vector \in \{0,1\}^m$), where each attribute is encoded as either 1 (included) or 0 (excluded):

$$change_vector = \begin{bmatrix} CA_1 \\ \vdots \\ CA_n \end{bmatrix} \quad (4.3)$$

In the third application step, the user selects the relevant methods (M) and digital tools (DT) to be considered. This selection is encoded likewise in binary form within a single vector ($MDT_vector \in \{1,9\}^{k+d}$), where a value of 1 indicates inclusion and 0 indicates exclusion:

$$MDT_vector = \begin{bmatrix} M_1 \\ \vdots \\ DT_d \end{bmatrix} \quad (4.4)$$

Each M and DT is characterized by a set of method attributes (MA) or digital tool attributes (DTA), which describe, for example, the effort required for its application or the type of activity it supports. This characterization is also stored in matrix form for M ($dmm_method \in \{1,9\}^{k \times r}$) and likewise for DT ($dmm_digital_tool \in \{1,9\}^{d \times o}$):

$$dmm_method = \begin{bmatrix} M_1 \rightarrow MA_1 & \cdots & M_1 \rightarrow MA_r \\ \vdots & \ddots & \vdots \\ M_k \rightarrow MA_1 & \cdots & M_k \rightarrow MA_r \end{bmatrix} \quad (4.5)$$

In addition, the correlation matrices developed in the first two phases of this study are incorporated as a central input to the methodology. These matrices capture the relationships between different attribute domains. As an example, the following matrix ($dmm_CA_PA \in \{1,9\}^{n \times m}$) shows the correlation between change attributes (CAs) and process attributes (PAs):

$$dmm_CA_PA = \begin{bmatrix} CA_1 \rightarrow PA_1 & \cdots & CA_1 \rightarrow PA_m \\ \vdots & \ddots & \vdots \\ CA_n \rightarrow PA_1 & \cdots & CA_n \rightarrow PA_m \end{bmatrix} \quad (4.6)$$

Analogous matrices exist for the correlations between CAs and MA/DTAs (denoted as $dmm_CA_MA_DTA \in \mathbb{N}_0^{n \times d}$) and between PAs and MA/DTAs (denoted as $dmm_PA_MA_DTA \in \mathbb{Z}_0^{m \times d}$).

With all initial matrices defined, the methodology proceeds to operational use. For each new change scenario, the user assigns values, ranging from 1 to 9, to all previously selected change attributes (i.e., those marked with 1 in the initial $change_vector$). This results in an updated, case-specific change vector.

The first computational step (CS1) involves multiplying this vector with the correlation matrices dmm_CA_PA and $dmm_CA_MA_DTA$ to derive change-specific relevance scores for both PAs and MAs/DTAs. The resulting vector for the process attributes is denoted as $R_change \in \mathbb{N}_0^m$:

$$\begin{aligned} \text{CS1 :} \\ R_change = \\ = dmm_CA_PA \bullet change_vector = \begin{bmatrix} cPA_1 \\ \vdots \\ cPA_m \end{bmatrix} \end{aligned} \quad (4.7)$$

An analogous vector is generated for *MAs/DTAs* and referred to as $R_change_MA_DTA \in \mathbb{N}_0^{r+o}$.

Based on the change-specific process attribute vector, the first outcome of the algorithm can now be derived, namely the identification of relevant process steps. In computational step 2 (CS2), the matrix $dmm_process$, which links each process step to its attributes, is multiplied by R_change . This yields the vector $process_final \in \mathbb{N}_0^p$, in which each entry represents the calculated relevance of a process step for the current change scenario:

$$\begin{aligned} \text{CS2 :} \\ process_final = dmm_process \bullet R_change = \begin{bmatrix} cPS_1 \\ \vdots \\ cPS_p \end{bmatrix} \end{aligned} \quad (4.8)$$

Higher values indicate greater relevance. Thresholds for excluding process steps can be defined individually by each company based on internal criteria.

The next objective is to determine which M&DTs are relevant for each process step in the context of a specific change. In computational step 3 (CS3), this is achieved by multiplying the matrix that links process steps to their attributes ($dmm_process$) with the correlation matrix that maps *PAs* to *MAs/DTAs*. This results in a relevance matrix ($R_process_MA_DTA \in \mathbb{N}_0^{px(r+o)}$) that indicates for each process step, which method or tool characteristics are generally suitable, independent of the specific change scenario:

$$\begin{aligned} \text{CS3 :} \\ R_process_MA_DTA = \\ dmm_process \bullet dmm_PA_MA_DTA = \\ = \begin{bmatrix} PS_1 \rightarrow MA_1 & \cdots & PS_1 \rightarrow DTA_d \\ \vdots & \ddots & \vdots \\ PS_p \rightarrow MA_1 & \cdots & PS_p \rightarrow DTA_d \end{bmatrix} \end{aligned} \quad (4.9)$$

At this point, the result is a matrix that indicates which *MAs/DTAs* are generally relevant for each process step. To incorporate the change-specific perspective, this matrix must now be weighted according to the specific change scenario. In computational step 4 (CS4), this is done through an element-wise multiplication with the change-specific attribute weights captured in $R_change_MA_DTA$. Since the element-wise (Hadamard) multiplication requires matching dimensions, the vector $R_change_MA_DTA$ is first transformed into an auxiliary matrix $T_change_MA_DTA \in \mathbb{N}_0^{px(r+o)}$, which replicates the vector across p rows, where p is the number of process

steps. This transformation ensures compatibility with the previously computed matrix of general M&DTs relevance per process step. The resulting weighted matrix ($cR_process_MA_DTA \in \mathbb{N}_0^{px(r+o)}$) reflects the change-specific importance of each of the *MAs/DTAs* for each individual step in the change process:

$$\begin{aligned} \text{CS4 :} \\ cR_process_MA_DTA = \\ R_process_MA_DTA \circ T_change_MA_DTA = \\ = \begin{bmatrix} PS_1cMA_1 & \cdots & PS_1cDTA_o \\ \vdots & \ddots & \vdots \\ PS_pcMA_1 & \cdots & PS_pcDTA_o \end{bmatrix} \end{aligned} \quad (4.10)$$

At this stage, the algorithm has computed the change-specific relevance of the *MAs/DTAs* for each process step. In the final computational step (CS5), these abstract attributes are linked to concrete M&DTs. To achieve this, the attribute profiles of all selected M&DTs are merged into a single matrix and transposed to align with the weighted relevance matrix. The resulting operation yields a final matrix ($M_DT_final \in \mathbb{N}_0^{px(k+d)}$) indicating for each process step, how well each specific M&DTs aligns with the requirements of the current change scenario:

$$\begin{aligned} \text{CS5 :} \\ M_DT_final = \\ cR_process_MA_DTA \bullet dmm_M_DTT = \\ = \begin{bmatrix} PS_1cM_1 & \cdots & PS_1cDT_d \\ \vdots & \ddots & \vdots \\ PS_pcM_1 & \cdots & PS_pcDT_d \end{bmatrix} \end{aligned} \quad (4.11)$$

As with the process steps, company-specific thresholds can be defined to determine when a method or digital tool is considered sufficiently relevant to be recommended by the approach. The sensitivity of the resulting process configurations and M&DTs recommendations, therefore, depends on the company-specific threshold values applied within the algorithm. Process steps or M&DTs are considered relevant only when their calculated relevance exceeds defined thresholds, which can be configured individually for each process step and each method or digital tool. This threshold-based logic allows companies to adapt the sensitivity of the results to their specific context and reduces the impact of minor rating variations, thereby increasing the robustness and practical applicability of the approach.

5 Discussion

This contribution introduced an approach that integrates expert-based correlations and algo-rithmic computation to support change-specific and company-individual MCM. By combining a hybrid Delphi study with a mathematical model, the approach provides the computational core to

derive tailored process configurations and suitable M&DTs for varying change scenarios. As with any complex methodology, the approach entails certain strengths, assumptions, and limitations that require critical reflection. Some of these aspects, particularly those related to the practical application and evaluation, have been further explored in adjacent work and are discussed in Sect. 5.1. Other conceptual and methodological considerations, such as of expert bias, data availability, and scalability, are addressed in Sect. 5.2.

5.1 Adjacent future work

5.1.1 Software implementation

A key limitation of the methodology in its raw form is the manual effort required for the configuration and the execution, particularly when adapting the method to new companies. Without digital support, applying the logic operationally would be time-consuming and prone to errors. This limitation is addressed through the development of a web-based application. The tool integrates the full logic of the CCMCM algorithm and provides a guided interface for users to configure their change processes, select relevant attributes, and assess upcoming changes. Based on the user input, the system generates tailored workflows with M&DTs recommendations. This implementation enables a scalable and reproducible application of the methodology, significantly reducing the need for manual calculations or interpretation. Moreover, it ensures the correct application of the logic and supports companies in continuously documenting and improving their change processes while also collecting change process data.

5.1.2 Industrial application and evaluation

Another potential limitation lies in the practical validity of the methodology. While the algorithmic model and correlation logic are theoretically grounded, their impact in real-world industrial environments was not tested within the scope of this paper. This is addressed in a separate evaluation study [42], which applies the complete methodology across three industrial use cases with varying levels of process maturity and data availability. In each case, process engineers use the method to evaluate current and upcoming changes. First results show that the method enables the omission of unnecessary process steps, improves transparency in tool selection, and facilitates communication with stakeholders. Importantly, users report improved confidence in their change planning decisions. These findings empirically support the practical value of the methodology and demonstrate its adaptability to different industrial contexts.

5.2 Critical reflection and outlook

Beyond the topics addressed in the adjacent work, several additional aspects require critical reflection to assess the robustness, generalizability, and practical relevance of the approach.

Expert dependence The correlation matrices at the core of the approach are based on subjective expert judgment. While the hybrid Delphi method offers a robust format, expert bias or inconsistency cannot be fully ruled out. To partially address this, the second phase of the study included a data-based verification. Still, data availability limited this to process steps, leaving M&DTs correlations unverified. Future industrial use and digital logging of M&DTs usage could allow for a more comprehensive validation.

Modeling simplifications The algorithm uses linear matrix operations, simple scales and additive logic to ensure interpretability and low complexity. While this improves usability and transparency, it also introduces simplifications. Real-world relationships may be non-linear or conditional. Future enhancements could explore more advanced mathematical techniques, such as multi-criteria optimization or fuzzy inference systems, while carefully managing complexity and usability (c.f. [43, 44]).

Scalability and transferability The approach can be scaled and transferred to different industrial settings, including environments with high product variability or modular production lines, by adapting process models, attributes, and threshold values to the specific context. However, transferability may be limited if company-specific processes or data are insufficiently formalized, which can affect the quality of the resulting configurations. Clear structuring of processes and attributes, as well as guided configuration support, can facilitate transfer and ensure consistent application across different industrial contexts.

Configurability vs. complexity The method is highly configurable, but this also increases the risk of inconsistent or biased setups. For instance, poor attribute selection could lead to suboptimal process suggestions. Structured onboarding processes, validation rules, or semi-automated recommendations might improve robustness and usability.

Future research could extend the proposed approach by integrating real-time monitoring and adaptive reconfiguration mechanisms. Instead of relying on static correlation matrices, dynamic updates could be enabled through feedback loops based on operational data, for example by leveraging sensor data or digital twins. Such an extension would allow the methodology to continuously adapt correlation strengths and recommendations to changing system states and operational conditions.

6 Conclusion

This paper addresses a central gap in current MCM research: the absence of a methodologically grounded and operationally usable approach to link change types, company-individual process models, and suitable M&DTs in a flexible and configurable way. Existing approaches tend to rely on fixed change classes, generalized processes, or descriptive M&DTs integration, making them insufficient for dynamic and company-specific change scenarios.

To close this gap, this contribution developed the essential components needed to implement such a methodology. First, a hybrid Delphi study was conducted, integrating expert assessments with GenAI-generated input to derive initial correlation matrices between change, process, and M&DTs attributes. Second, selected correlations were verified using real-world industrial data, enhancing the validity of the results. Third, an algorithm was developed that transforms these structured inputs into operational output: a change-specific process configuration enriched with tailored M&DTs recommendations. By combining expert reasoning, empirical evidence, and computational logic, the paper enables a more adaptive, traceable, and efficient way of configuring change processes. This marks a shift from static, one-size-fits-all MCM practices to a modular and data-supported decision logic that is tailored to the nature of the change and the company's individual context.

While this contribution lays the methodological and computational foundation, future work will focus on several other directions. Most immediately, the methodology should be applied in broader industrial contexts to assess generalizability across sectors, company sizes, and change types. This includes testing in settings with lower process maturity or higher variability in change frequency. In parallel, expanding the available datasets, especially regarding M&DTs usage, will enable a deeper empirical validation. Another potential lies in refining the mathematical model. Ultimately, this work opens the path towards a more data-informed, flexible, change-specific, and company-individual approach to managing MCs.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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